Zach Teafatiller

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Education

University of Colorado Boulder, College of Engineering and Applied Science Boulder, CO

Major: (M.S.) Mechanical Engineering - GPA: **3.76/4.0** *Expected Graduation: May 2028*

Minor: Engineering Management

Relevant Coursework: Statics, Thermodynamics, Materials Science, Mechanics of Solids, Fluid Mechanics, Dynamics

Engineering Experience

Assistant Project Manager, Ride Engineering Competition **October 2025 - April 2026**

- Designed and manufactured a 1:50 scale themed attraction with an interdisciplinary team capable of elevation and rotation
- Fabricated and assembled a scissor lift capable of 1 foot of elevation while staying within scale and packaging constraints
- Coordinated between 5-10 person sub-teams to integrate designs and maintain project milestones
- Documented design reviews and contributed to deliverables submitted for competition judging, including ASTM F2291 compliance documentation

Pencil Sharpener CAD Project **October 2025 - December 2025**

- Reverse-Engineered a hand crank pencil sharpener to create a model of its planetary gear motion and internal components
- Performed measurements of a physical product to create a SolidWorks assembly with gear mates and irregular geometry
- Developed a full CAD package consisting of part models, subassemblies, and fully dimensioned engineering drawings

Additional Work Experience

Treasurer, CU Theme Park Engineering and Design Club **April 2026 - Present**

- Manage and track funding sources for club projects, events, and engineering competitions
- Oversee a budget of $5000 in Excel, outline Bills of Materials and track expenditures to ensure budget compliance
- Write and present funding proposals to university leadership to secure financial support for organizational needs
- Collaborate with other officers to plan events, meetings, and outreach opportunities in the local community

Team Member/Baker, Panera Bread **March 2024 - August 2025**

- Managed daily bake schedules of ~20 pastry types to optimize for efficiency while meeting quality standards
- Trained new employees on baking procedures, food safety, and company standards
- Coordinated with team members across shifts to maintain production continuity and meet daily demand
- Assisted customers and collaborated with front-of-house staff in a fast-paced service environment

Skills and Qualifications

CAD Modeling & Programming: SolidWorks (CSWA), Arduino, MATLAB, C, Python

Fabrication: 3D Printing, Laser Cutting, Turning, Milling

Documentation: Microsoft Suite (Excel, Word, PowerPoint), ASTM F2291

Societies & Affiliations: Theme Park Engineering and Design Club, American Society of Mechanical Engineers, American Society for Engineering Management